# COM3502-4502-6502
# SPEECH PROCESSING

## Lecture 09

## Linear Filters



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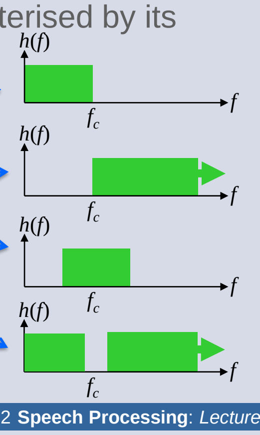
## Filters

- A ‘filter’ modifies the properties of a signal
- A filter’s effect can be analysed in the time domain or in the frequency domain
- A filter is typically characterised by its ‘frequency response’
- Types of filter …
  - low-pass
  - high-pass
  - band-pass
  - band-stop



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# Filters

- The ‘frequency response’ of a *low-pass* or a *high-pass* filter is characterised by …
  - the ‘cut-off frequency’ $f_c$ (*Hz*)
  - the ‘roll-off rate’ (*dB/octave*)
- A ‘first-order filter’ attenuates at a roll-off rate of 6 dB/octave
- The frequency response of a *band-pass* filter is characterised by
  - the ‘centre frequency’ $f_c$ (*Hz*)
  - the ‘bandwidth’ (*Hz*) or ‘Q’ (=*cf/bw*)
- When a signal passes through a linear filter, its spectrum is *multiplied* by the frequency response of the filter



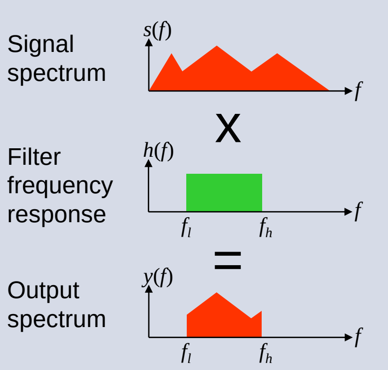




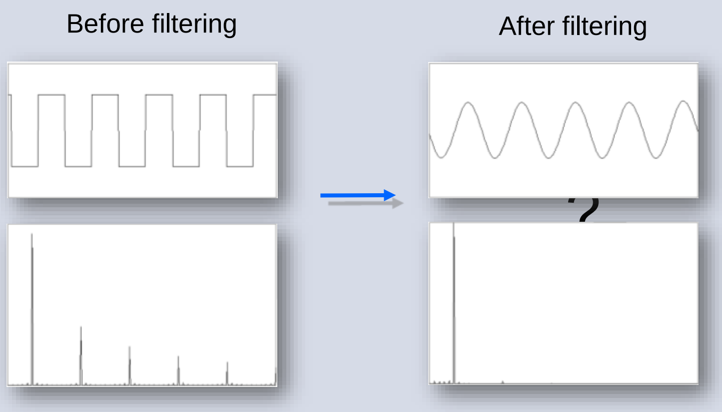


# Filters

Signal $s(f)$ → Filter $h(f)$ → Filtered Signal $y(f) = h(f)\,s(f)$



# Example of a Filter



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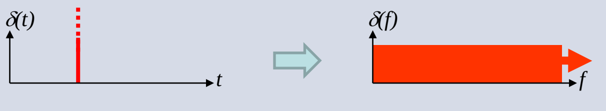
# Impulse Response

- An ‘impulse’ signal is an instantaneous pulse of energy
- An idealised impulse signal has a *flat* frequency spectrum (*i.e. equal energy at all frequencies*)




- Hence the ‘impulse response’ of a filter has a spectrum that is *equal* to its frequency response
- Note that white noise *also* has a flat spectrum



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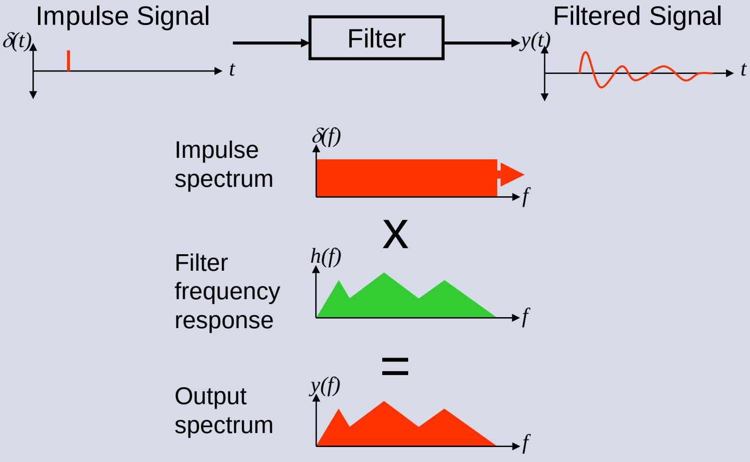
# Impulse Response

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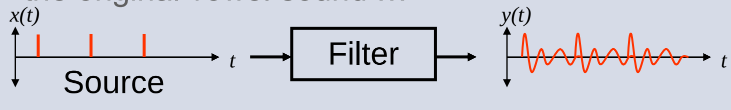
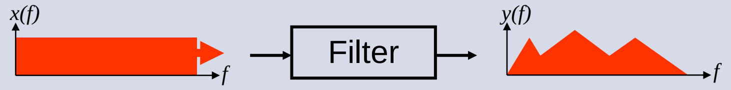
# Source-Filter Modelling

- If we design a filter with a frequency response equal to the spectrum of a vowel sound and input a sequence of impulses, we will generate the original vowel sound …




- Also, if we design a filter with the frequency response of an unvoiced fricative and input a white noise signal, we will generate the original fricative sound …

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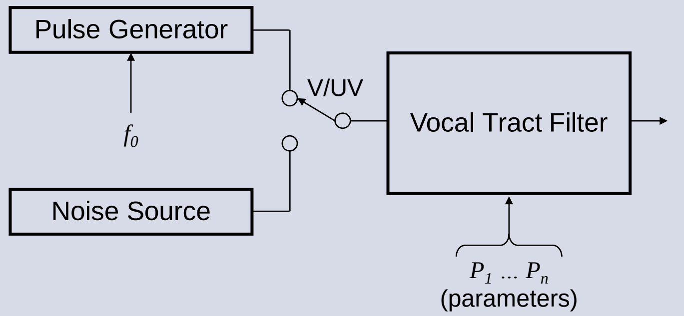
# Source-Filter Model of Speech




- We will learn this in much more detail when we discuss linear predictive coding (LCP) in a later lecture

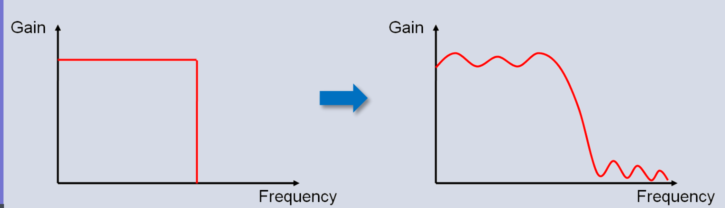




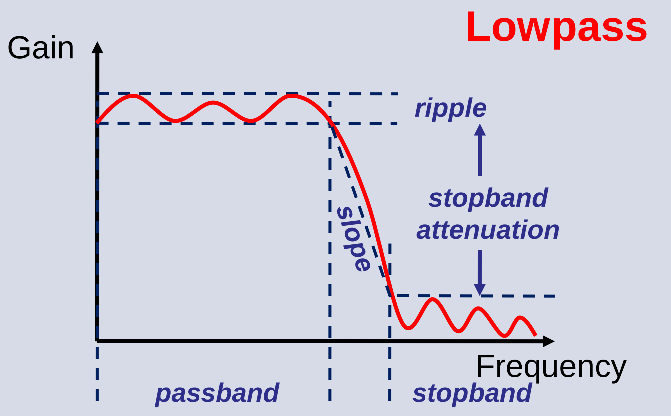
# Idealised *vs.* Practical Filters

- Filters are often described in idealised terms, e.g. as ‘brickwall’ or ‘piecewise linear’ filters
- In reality, practical filters approximate the idealised forms

*Idealised ‘Brickwall’ Filter* *Practical Filter*



# Practical Filters



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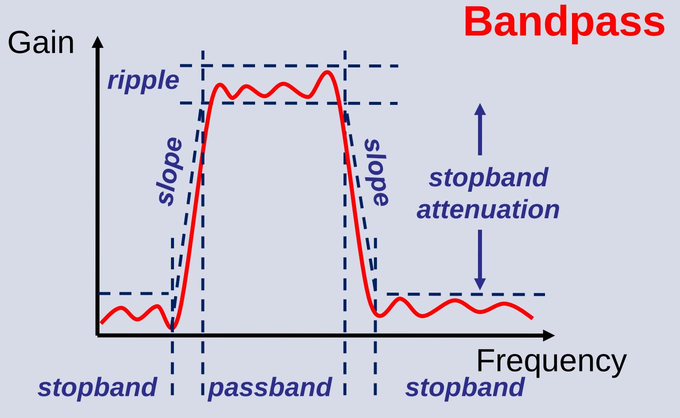
# Practical Filters



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# Filter Design

- The behaviour of a filter is defined by the number and the location of its ‘poles’ and ‘zeros’
(*as we saw in Lecture #7 – Z-Transform*)
- The complexity of a filter is often characterised by its ‘order’
- In a digital filter, order is defined as the number of *past* (*input or output*) values that are involved in the calculation
- For example, a filter that takes two past values is termed a ‘second-order filter’
- Filter order can thus be related to the number of poles and zeros
- Hence, the degree to which a practical filter approximates the idealised form is a function of its order
(*i.e. the number of poles and zeros*)

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# General Difference Equation

- The general difference equation may include any number of past inputs and outputs …

$$y[k] = a_1 y[k-1] + a_2 y[k-2] + \ldots + a_p y[k-p] + \ldots$$
$$+ b_0 x[k] + b_1 x[k-1] + b_2 x[k-2] + \ldots + b_q x[k-q]$$

- Taking Z transforms …

$$Y(Z) = Y(z)\sum_{i=1}^{p} a_i z^{-i} + X(z)\sum_{i=0}^{q} b_i z^{-i}$$

$$H(z) = \frac{\sum_{i=0}^{q} b_i z^{-i}}{1 - \sum_{i=1}^{p} a_i z^{-i}}$$

zeros

poles

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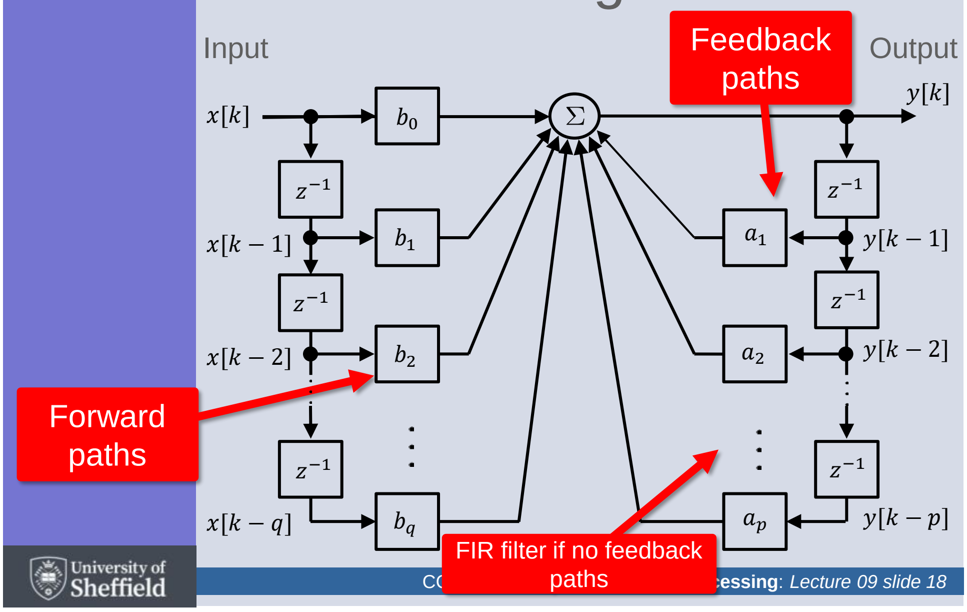
# General Digital Filter



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# General Digital Filter

- A very wide range of transfer functions (*and associated frequency responses*) may be obtained by appropriate choice of the filter parameters
- $p$ = number of poles, $q$ = number of zeros
- For the case $p = 0$, the resulting non-recursive filter is said to be a 'finite impulse response - FIR' filter
- If $p > 0$, the resulting recursive filter is said to be an 'infinite impulse response - IIR' filter



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# 'FIR' Filters

- Finite impulse response (*FIR*) filters express each output sample as a weighted sum of the last $N$ inputs (*where* $N$ *is the order of the filter*)
- Advantages
  - inherently stable since they don't use feedback (*i.e. only zeros*)
  - the coefficients are usually symmetrical, hence the phase response is linear and signals of all frequencies are delayed equally
  - overflow is straightforward to avoid
  - generally easier to design than IIR filters
- Disadvantages
  - may require significantly more processing and memory resources than the equivalent IIR filter
  - often require a much higher filter order than IIR filters
  - delay can be much greater than for an equivalent IIR filter





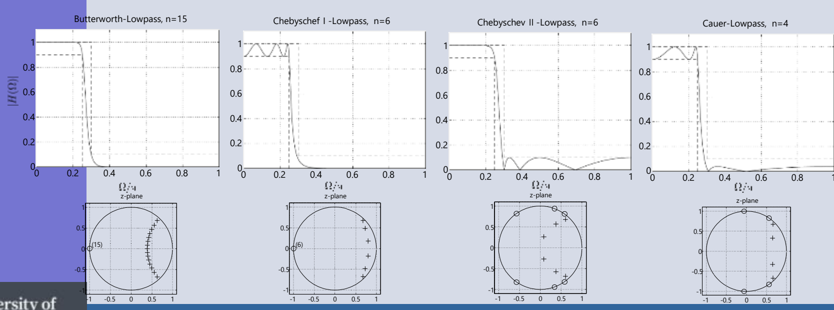
# 'IIR' Filters

- Infinite impulse response (*IIR*) filters are the digital counterpart to analogue filters

  … *because they contain an internal state, and the output and the next internal state are determined by the previous inputs and outputs*
- Advantages
  - normally require less computing resources than an FIR filter of similar performance
- Disadvantages
  - due to the feedback, high-order IIR filters may have problems with instability, arithmetic overflow and limit cycles
  - careful design is required to avoid such pitfalls
  - the time delay through such a filter is frequency-dependent (*since the phase shift is inherently a non-linear function of frequency*)

# ‘IIR’ Filters

- IIR filters all approximate the ideal *brick-wall* filter …
  - Butterworth
  - Chebyshev (*types I and II*)
  - Elliptic (Cauer)
  - Bessel

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# ‘IIR’ Filters

- IIR filters all approximate the ideal *brick-wall* filter …
  - Butterworth
  - Chebyshev (*types I and II*)
  - Elliptic
  - Bessel
- High-order IIR filters can easily become unstable
- This is much less of a problem with first and second-order filters
- 2nd-order IIR filters are often called ‘biquads’
- Higher-order filters are typically implemented as a cascade of ‘biquad sections’

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# ‘Biquad’ Filters

- A ‘biquadratic’ (*biquad*) filter is a 2nd-order recursive linear IIR filter, containing *two* poles and *two* zeros
- The name refers to the fact that its Z domain transfer function is the ratio of two *quadratic* functions …

$$H(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 + a_1 z^{-1} + a_2 z^{-2}}$$

- This is derived from the following difference equation …

$$y[k] = b_0 x[k] + b_1 x[k-1] + b_2 x[k-2] - a_1 y[k-1] - a_2 y[k-2]$$

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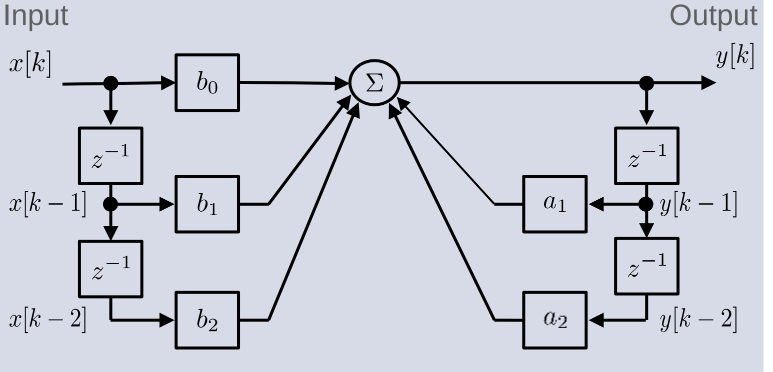
# ‘Biquad’ Filters




$$y[k] = b_0 x[k] + b_1 x[k-1] + b_2 x[k-2] - a_1 y[k-1] - a_2 y[k-2]$$

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# Filters in Speech Processing

- General filtering
  - notch filtering (*to remove interference and noise*)
  - pre-emphasis (*to equalise the frequency response*)
- Modelling speech production
  - the source-filter model (*Lecture 'Linear Prediction'*)
- Modelling the auditory system
  - filter-bank analysis (*Lecture 'Hearing'*)

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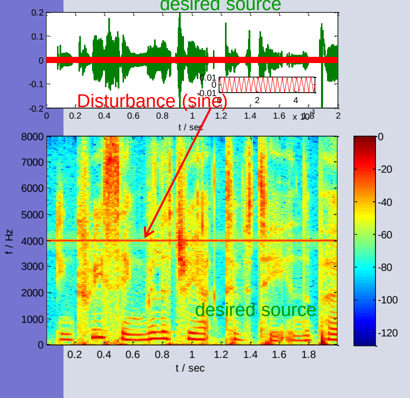
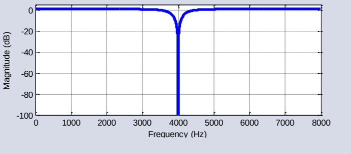
# Tone Removal by Notch Filtering




Solution: e.g. notch filer (stationary filter)

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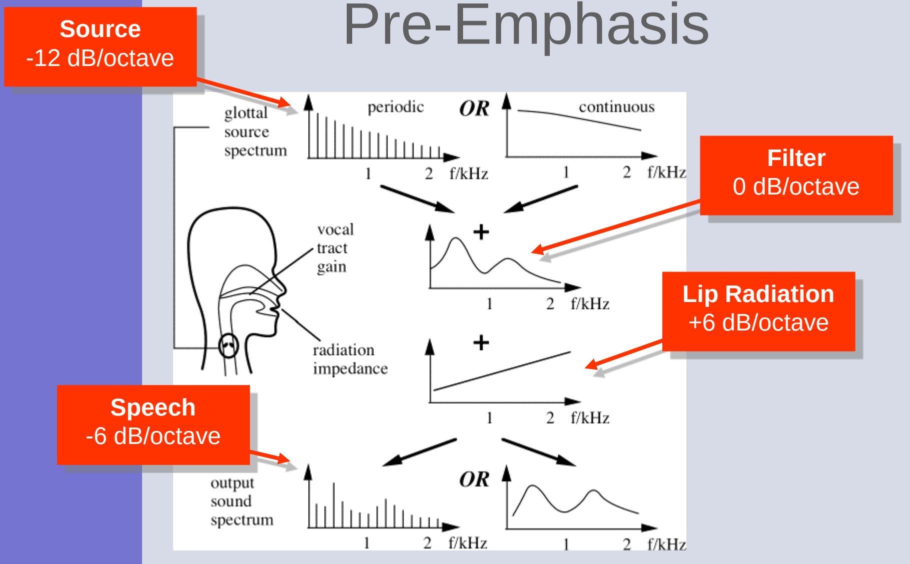
# Pre-Emphasis




http://www.phys.unsw.edu.au/jw/voice.html





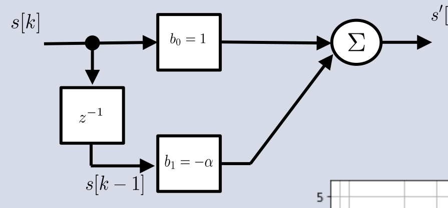
# Pre-Emphasis

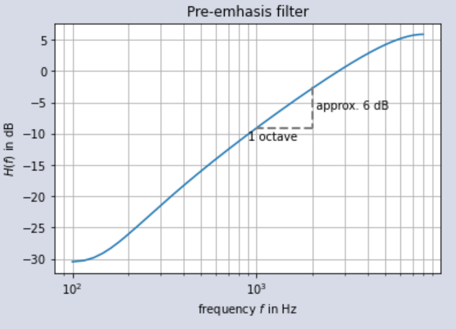
- Averaged over time, speech has an overall spectral slope of ~ -6 dB/octave
- This means that there can be a large amplitude difference between the lowest *(~50 Hz*) and highest (*~8 kHz*) frequency components
- E.g. over 8 octaves there will be a (6x8) 48 dB difference (*which is comparable to the dynamic range from the quietest to the loudest speech sounds*)
- This can make processing difficult (*especially for functions such as spectral peak picking, i.e. formant tracking*)
- It is therefore usual to 'pre-emphasise' a speech signal by giving it a +6 dB/octave lift

# Pre-Emphasis Filter

Input

Output



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# Auditory Filters

- Filterbanks associated with the DFT/FFT have constant bandwidths and are centered at uniformly spaced locations along the frequency axis
- To better model the frequency response characteristics of the human ear, many researchers use filters inspired by the auditory system
- These typically have *non-uniform* bandwidths and *non-uniform* spacing of center frequencies
- The most well-known auditory filter is the 'gammatone'
- This filter has an impulse response that is the product of a gamma distribution and sinusoidal tone

**Patterson, R. D., & Moore, B. C. J. (1986). Auditory filters and excitation patterns as representations of frequency resolution. In B. C. J. Moore (Ed.), *Frequency Selectivity in Hearing* (pp. 123-177). London: Academic Press Ltd.**



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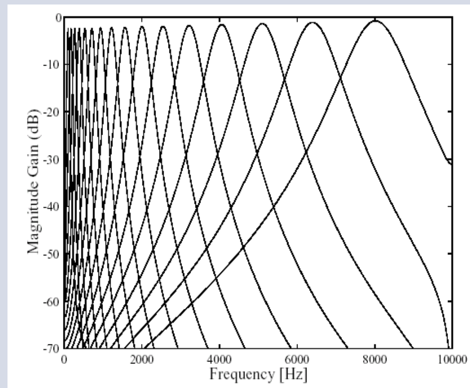
# Gammatone filterbank

Compare to lecture L11 (Hearing)

- Each position on the basilar membrane simulated by a gammatone filter with appropriate centre frequency and bandwidth.
- Smaller number of filters as for spectrograms
- variation in bandwidth with frequency (unlike Fourier analysis).








# This slide set has covered …

- (Piecewise linear) filter examples
- Idealised *vs.* practical filters
- Filter design
- Second-order filters
- The general difference equation
- FIR and IIR filters
- The ‘biquad’ filter
- Filters in speech processing
- The ‘gammatone’ filter

# Any Questions ?

*Raise during lecture or post in the Blackboard Discussion group*

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# Next time …

Speaking

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